Module 0: Introduction

Regex (& Testing)

Two of my favorite things combined - like Reese's!

1.

Regex

More like “not very regular” expressions… ba-dum tss…

Regex 101

Definition

A regex (**regular expression**) defines a pattern for matching text values.

- Regexes allow searching and validating text (esp. user input).
  - numbers, emails, file paths
- Also used for more advanced extraction/replacement of values.
  - “no swearing on my server!”

Regex is a Language!

Regex is a **Domain Specific Language (DSL)** that makes string matching easier.

- It has a grammar that provides a specification for valid regexes.
- It has a parser/compiler (engine).
- They’re not all-powerful, however.
  - Do not parse HTML with regex!
  - To be continued…

Regex Crash Course

Concatenation, Alternation, Character Classes, Quantifiers, Capturing Groups, Escaping

Java Regex Cheat Sheet

2.

Testing

~~*“Write tests because it looks good in interviews!”*~~

Why Test?

Popular Reasons

Testing is like, the *thing*, y'know?

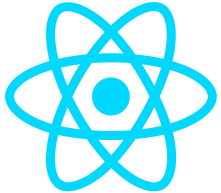
- Helps verify code is correct.
  - And *stays* correct.
- Makes development easier.
  - Especially within groups!
- My boss/company requires it.
  - Also we require 90% coverage!
    - Which is measured by lines :/

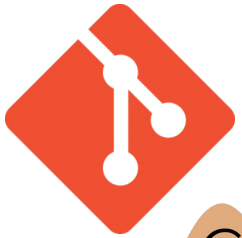
Better* Reasons

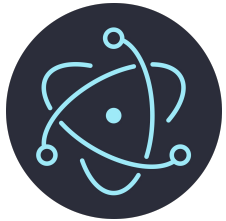
Testing aids the "hard" parts of software.

- Forces you to consider edge cases.
  - Happy path only leads to sad path.
- Self-document intended behavior.
  - "Any undocumented behavior is indistinguishable from a potential bug"
- Ensure dependencies work as expected
  - Libraries may have breaking changes!
  - Machine/browser/environment differences.

npx create-react-app







Testing Methodology

Test Coverage

Coverage is a metric for how well test cases "cover" the behavior of code.

- Helpful for identifying untested behavior and edge cases.
- Provides an estimate of how well the software is tested.
- Must consider coverage in context!
  - Adding coverage has costs too.
  - 90% coverage could mean 0% of the 10% that's actually valuable.

Equivalence Partitioning

Separate test input into categories that are expected to behave the same (per spec).

- Testing all inputs is often unfeasible.
  - `f(int)`: 2^32 integer values!
  - `f(int, int)`: 2^32 * 2^32 = 2^64
  - add in classes, state, side effects, etc…
- Instead, test a subset of values that cover all **equivalence classes**, i.e. unique behaviors.
  - Start with a baseline test (bare minimum)
  - Add variations on the baseline to cover new eq. classes with *minimal changes*.

FizzBuzz Equivalence Classes

Divisible by 3 <u>only</u>	3, 6, 9, 12, ~~15~~, 18, …	"Fizz"
Divisible by 5 <u>only</u>	5, 10, ~~15~~, 20, 25, 35, …	"Buzz"
Divisible by 3 <u>and</u> 5	15, 30, 45, 60, 75, …	"FizzBuzz"
All other numbers	1, 2, ~~3~~, 4, ~~5~~, ~~6~~, 7, 8, …	number

Edge Cases

Definitions

Edge cases are specific equivalence classes (or often just values) that have distinct behavior from others.

- Typically applies to a smaller set of values that is less likely to occur in "normal" programs.
- Also applies to "combinations" of values that lead to unique cases.
  - multiply two negatives = postive

Identification

Edge cases often fall into a few categories:

- Values with unique behavior for the type.
  - null, 0, empty strings/lists
- Boundary values: specific values at the edge of valid/invalid input, typically ranges.
- Error cases: known error conditions.
  - invalid user input, IO exceptions
- Extreme input: input intentionally designed to be difficult, tricky, or even malicious.

Also, look for specific call-outs in specifications or control flow variations in implementations.

FizzBuzz Edge Cases

`0, 1`	Boundary values (lower bound) 0/1 are also common edge cases (why?)
`100, 101`	Boundary values (upper bound)
`-1`	Boundary values (negative numbers) -1 is also a common edge case (why?)
`int min/max`	Boundary values (lower bound) INT_MIN is an interesting edge case (why?)

3.

Regex Testing

It’s time for the first assignment!

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